

MATRICES (1 + 3 + 5 = 9m)

1 marker : Definitions or construction of matrix

3 marker : Find values (p, q, r, x, y, z)

: $A = P + Q$ (sum of symmetric & skew symmetric matrices)

: A^{-1} by elementary transformation

5 marker : Proofs on properties

$$(AB)' = B'A'$$

$$A(BC) = (AB)C$$

$$A(B+C) = AB+AC$$

Definition: Matrix is a rectangular arrangement of elements in rows and columns.

Horizontal elements form rows and vertical elements form columns.

Matrices are represented by capital alphabets A, B, C, ... and the elements are enclosed in brackets like $[]$, $()$, $\langle \rangle$, $\| \|$.

General form of a matrix:

A_{mn} : matrix A with m rows and n columns.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{i1} & a_{i2} & a_{i3} & \dots & a_{in} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix} \quad \begin{array}{l} a_{ij} - \text{element present} \\ \text{in } i^{\text{th}} \text{ row and } j^{\text{th}} \\ \text{column.} \end{array}$$

$m \times n$

1) **Row matrix:** A matrix which has a single row and is of order " $1 \times n$ ".

$$A = [0 \ 1 \ -1 \ 3 \ 5]_{1 \times 5}$$

- 2] Column matrix : A matrix which has a single column and is of order " $m \times 1$ ".

$$A = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}_{3 \times 1}$$

- 3] Zero/ Null/ Void matrix ('0') : A matrix of any order in which all the elements are equal to zero.

$$O = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{2 \times 3}$$

- 4] Rectangular matrix : A matrix in which no. of rows is not equal to no. of columns i.e, $m \neq n$.

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & -1 & 2 \end{bmatrix}_{2 \times 4}$$

- 5] Square matrix : A matrix in which no. of rows is equal to no. of columns i.e, $m = n$.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}_{2 \times 2} \rightarrow 2^{\text{nd}} \text{ order matrix}$$

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}_{3 \times 3} \rightarrow 3^{\text{rd}} \text{ order matrix}$$

NOTE : Principal diagonal

$$A = \begin{bmatrix} 1 & 4 & 5 \\ -2 & 2 & 0 \\ -5 & 7 & 3 \end{bmatrix} \rightarrow \text{pd}$$

The diagonal in a matrix from left top to right bottom is called its principal diagonal. (The other diagonal is not considered for any property).

- 6] Diagonal matrix: It is a square matrix in which all the elements are equal to zero except the principal diagonal elements.

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

- 7] Scalar matrix: It is a diagonal matrix in which all the principal diagonal elements are equal.

$$A = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

- 8] Identity / Unit matrix (I): It is a scalar matrix in which all the principal diagonal elements are equal to 1 and is represented by "I".

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

NOTE: Transpose of a matrix A is obtained by interchanging rows and columns of matrix A and is denoted by A' .

$$A = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \\ 7 & 8 & 9 \end{bmatrix} \quad A' = \begin{bmatrix} 1 & 2 & 7 \\ 3 & 4 & 8 \\ 5 & 6 & 9 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 3 & 5 & 7 \\ 0 & 1 & 2 & 3 \end{bmatrix} \quad B' = \begin{bmatrix} 1 & 0 \\ 3 & 1 \\ 5 & 2 \\ 7 & 3 \end{bmatrix}$$

If order $(A) = m \times n$ then order $(A') = n \times m$

** $(AB)' = B'A'$ – Demorgan's Law

If element of A is a_{ij} then element of A' becomes a_{ji}

9] Symmetric matrix: A matrix in which $a_{ij} = a_{ji}$ i.e., $A' = A$

$$A = \begin{bmatrix} 3 & 2 & 7 \\ 2 & 4 & -1 \\ 7 & -1 & 5 \end{bmatrix} \quad A' = \begin{bmatrix} 3 & 2 & 7 \\ 2 & 4 & -1 \\ 7 & -1 & 5 \end{bmatrix}$$

10] Skew Symmetric matrix: A matrix in which $a_{ij} = -a_{ji}$ i.e., $A' = -A$ and all the principal diagonal elements are equal to zero.

$$A = \begin{bmatrix} 0 & 1 & -7 \\ -1 & 0 & 5 \\ 7 & -5 & 0 \end{bmatrix} \quad A' = \begin{bmatrix} 0 & -1 & 7 \\ 1 & 0 & -5 \\ -7 & 5 & 0 \end{bmatrix} = -\begin{bmatrix} 0 & 1 & -7 \\ -1 & 0 & 5 \\ 7 & -5 & 0 \end{bmatrix} = -A$$

11] Triangular matrix:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 2 & 0 \\ 5 & 6 & 3 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 4 & 6 \\ 0 & 2 & 5 \\ 0 & 0 & 3 \end{bmatrix}$$

lower triangular matrix

upper triangular matrix

Equality of matrices:

Two matrices $A = [a_{ij}]$ and $B = [b_{ij}]$ are said to be equal if

(i) They are of same order

(ii) Each element of A is equal to the corresponding element of B i.e., $a_{ij} = b_{ij}$ for all i and j .

$$\begin{bmatrix} x & 3 \\ 0 & y \end{bmatrix} = \begin{bmatrix} 1 & a \\ b & 2 \end{bmatrix}$$

$$\Rightarrow x = 1, a = 3 \\ b = 0, y = 2$$

EXERCISE 3.1

1] In the matrix $A = \begin{bmatrix} 2 & 5 & 19 & -7 \\ 35 & -2 & 5/2 & 12 \\ \sqrt{3} & 1 & -5 & 17 \end{bmatrix}$ write:

(i) order of matrix = 3×4

(ii) no. of elements = 12

iii] $a_{13} = 19$ $a_{24} = 12$
 $a_{21} = 35$ $a_{23} = 5/2$
 $a_{33} = -5$ $a_{34} = 17$

**2] If a matrix has 24 elements, what are the possible orders it can have? What, if it has 13 elements.

Sol] Given: 24 elements

Possible orders: 1×24 , 24×1
 2×12 , 12×2
 3×8 , 8×3
 4×6 , 6×4

Given: 13 elements

Possible orders: 1×13 , 13×1

**3] If a matrix has 18 elements, what are the possible orders it can have? What, if it has 5 elements.

Sol] Given: 18 elements

Possible orders: 1×18 , 18×1
 2×9 , 9×2
 3×6 , 6×3

Given: 5 elements

Possible orders: 1×5 , 5×1

ACHIEVER

4] Construct 2×2 matrix $A = [a_{ij}]$ whose elements are given by,

(i) $a_{ij} = \frac{(i+j)^2}{2}$

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} 2 & 9/2 \\ 9/2 & 8 \end{bmatrix}$$

(ii) $a_{ij} = \frac{i}{j}$

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} 1 & 1/2 \\ 2 & 1 \end{bmatrix}$$

(iii) $a_{ij} = \frac{(i+2j)^2}{2}$

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} 9/2 & 25/2 \\ 8 & 18 \end{bmatrix}$$

5] Construct a 3×4 matrix, whose elements are given by:

(i) $a_{ij} = \frac{1}{2} |-3i+j|$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix} = \begin{bmatrix} 1 & 1/2 & 0 & 1/2 \\ 5/2 & 2 & 3/2 & 1 \\ 4 & 7/2 & 3 & 5/2 \end{bmatrix}$$

(ii) $a_{ij} = 2i-j$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 & -2 \\ 3 & 2 & 1 & 0 \\ 5 & 4 & 3 & 2 \end{bmatrix}$$

6] Find the values of x, y, z from the following equations.

$$(i) \begin{bmatrix} 4 & 3 \\ x & 5 \end{bmatrix} = \begin{bmatrix} y & z \\ 1 & 5 \end{bmatrix}$$

$$\Rightarrow x=1, y=4, z=3$$

$$(ii) \begin{bmatrix} x+y & 2 \\ 5+z & xy \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 5 & 8 \end{bmatrix}$$

$$x+y=6 \quad 5+z=5 \Rightarrow \boxed{z=0}$$

$$y=6-x$$

$$y=6-2 \text{ (or) } 6-4$$

$$\boxed{y=4, 2}$$

$$xy=8$$

$$x(6-x)=8$$

$$6x-x^2-8=0$$

$$x^2-6x-8=0$$

$$(x-4)(x-2)=0$$

$$\boxed{x=2, 4}$$

$$\therefore x=2, y=4, z=0$$

$$x=4, y=2, z=0$$

$$(iii) \begin{bmatrix} x+y+z \\ x+z \\ y+z \end{bmatrix} = \begin{bmatrix} 9 \\ 5 \\ 7 \end{bmatrix}$$

$$\Rightarrow x+y+z=9$$

$$x+z=5$$

$$\Rightarrow y+5=9$$

$$\boxed{y=4}$$

$$y+z=7$$

$$p=z=7-4$$

$$\boxed{z=3}$$

$$x+z=5$$

$$x=5-3$$

$$\boxed{x=2}$$

7] Find the value of a, b, c, d :

$$\begin{bmatrix} a-b & 2a+c \\ 2a-b & 3c+d \end{bmatrix} = \begin{bmatrix} -1 & 5 \\ 0 & 13 \end{bmatrix}$$

$$a-b=-1$$

$$2a+c=5$$

$$3c+d=13$$

$$2a-b=0$$

$$a-b=-1$$

$$c=5-2$$

$$d=13-9$$

$$-a=-1-1$$

$$\boxed{c=3}$$

$$\boxed{d=4}$$

$$\boxed{a=1}$$

$$\boxed{b=2}$$

ACHIEVER

8] $A = [a_{ij}]_{m \times n}$ is a square matrix if
 (a) $m < n$ (b) $m > n$ (c) $m = n$ (d) none

9] Which of the given values of x, y make the following pair of matrices equal

$$\begin{bmatrix} 3x+7 & 5 \\ y+1 & 2-3x \end{bmatrix} = \begin{bmatrix} 0 & y-2 \\ 8 & 4 \end{bmatrix}$$

(a) $x = \frac{-1}{3}, y = 7$ (b) $y = 7, x = \frac{-2}{3}$
 (c) $x = \frac{-1}{3}, y = \frac{-2}{3}$ (d) $x = \frac{-1}{3}, y = \frac{-2}{3}$

$$3x+7=0$$

$$2-3x=4$$

$$\Rightarrow x = \frac{-7}{3}$$

$$x = \frac{-2}{3}$$

x has two different values which is not valid.

**10] The no. of all possible matrices of order 3×3 with each entry 0 or 1 is

(a) 27 (b) 18 (c) 81 (d) 512

Ans] Every entry has two possibilities (0 or 1)

No. of elements = 9

$\therefore$ no. of all possible matrices = $2^9 = 512$

*] Consider a 2×2 matrix whose elements

$$(i) a_{ij} = \frac{(i-2j)^2}{2} \quad (ii) a_{ij} = \frac{|2i-3j|}{2}$$

*] Construct a 2×3 matrix such that $a_{ij} = e^{ix} \sin jx$
 Given: $a_{ij} = e^{ix} \sin jx$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} = \begin{bmatrix} e^x \sin x & e^x \sin 2x & e^x \sin 3x \\ e^{2x} \sin x & e^{2x} \sin 2x & e^{2x} \sin 3x \end{bmatrix}$$

(b) $a_{ij} = |i-j|$ $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 1 \end{bmatrix}$

*] Construct a 4×3 matrix whose elements are given by

(a) $a_{ij} = \begin{cases} i^2 & \text{if } i < j \\ \frac{i}{j} & \text{if } i = j \\ j^2 & \text{if } i > j \end{cases}$

$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \\ a_{41} & a_{42} & a_{43} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 4 \\ 1 & 4 & 1 \\ 1 & 4 & 9 \end{bmatrix}$

(b) $a_{ij} = \begin{cases} ij-j & \text{if } i < j \\ \frac{i}{j} & \text{if } i = j \\ ij-i & \text{if } i > j \end{cases}$

$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \\ a_{41} & a_{42} & a_{43} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 3 \\ 0 & 3 & 1 \\ 0 & 4 & 8 \end{bmatrix}$

Operation of matrices:

Addition of matrices:

$A = \begin{bmatrix} \sqrt{3} & 1 & -1 \\ 2 & 3 & 0 \end{bmatrix}$ $B = \begin{bmatrix} 2\sqrt{5} & -1 \\ -2 & 3 & 1/2 \end{bmatrix}$

Find (i) $A + 2B$ (ii) $2A - B$

(ii) $\begin{bmatrix} 2\sqrt{3} & 2 & -2 \\ 4 & 6 & 0 \end{bmatrix} - \begin{bmatrix} 2\sqrt{5} & -1 \\ -2 & 3 & 1/2 \end{bmatrix} = \begin{bmatrix} 2(\sqrt{3}-1) & 2-\sqrt{5} & -3 \\ 6 & 3 & -1/2 \end{bmatrix}$

(i) $\begin{bmatrix} \sqrt{3} & 1 & -1 \\ 2 & 3 & 0 \end{bmatrix} + \begin{bmatrix} 4 & 2\sqrt{5} & 2 \\ -4 & 6 & 1 \end{bmatrix} = \begin{bmatrix} 4+\sqrt{3} & 1+2\sqrt{5} & 1 \\ -2 & 9 & 1 \end{bmatrix}$

Achiever

To add or subtract two matrices, the matrices have to be of same order.

We add or subtract the corresponding elements of the two matrices.

Multiplication of a matrix by a scalar k :

To multiply a matrix A by a scalar k , every element of A is multiplied by k i.e., $kA = k[a_{ij}]_{m \times n}$
 $= [k(a_{ij})]_{m \times n}$

Negative of a matrix:

Negative of a matrix A is $-A$.

$-A = (-1)A$, in which sign of every element of A is changed.

Properties of Matrix Addition:

- i) Commutative law: $A + B = B + A$
- ii) Associative law: $A + (B + C) = (A + B) + C$
- iii) Existence of Identity: Null matrix O is identity element under matrix addition
 $\therefore A + O = O + A = A$
- iv) Existence of Inverse: $-A$ is additive inverse of $+A$
 $\therefore -A + A = +A - A = O$

*] If $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 3 & -1 & 3 \\ -1 & 0 & 2 \end{bmatrix}$ find $2A - B$

$$\begin{bmatrix} 2 & 4 & 6 \\ 4 & 6 & 2 \end{bmatrix} - \begin{bmatrix} 3 & -1 & 3 \\ -1 & 0 & 2 \end{bmatrix} = \begin{bmatrix} -1 & 5 & 3 \\ 5 & 6 & 0 \end{bmatrix}$$

*] If $A = \begin{bmatrix} 8 & 0 \\ 4 & -2 \\ 3 & 6 \end{bmatrix}$ & $B = \begin{bmatrix} 2 & -2 \\ 4 & 12 \\ -5 & -1 \end{bmatrix}$ find matrix X

such that $2A + 3X = 5B$

$$3X = 5B - 2A$$

$$= \begin{bmatrix} 10 & -10 \\ 20 & 10 \\ -25 & 5 \end{bmatrix} + \begin{bmatrix} -16 & 0 \\ -8 & 4 \\ -6 & -12 \end{bmatrix}$$

$$= \begin{bmatrix} -6 & -10 \\ 12 & 14 \\ -31 & -7 \end{bmatrix}$$

$$X = \begin{bmatrix} -2 & -10/3 \\ 4 & 14/3 \\ -31/3 & -7/3 \end{bmatrix}$$

*] Find x and y if $x + y = \begin{bmatrix} 5 & 2 \\ 0 & 9 \end{bmatrix}$, $x - y = \begin{bmatrix} 3 & 6 \\ 0 & -1 \end{bmatrix}$

$$\textcircled{1} + \textcircled{2} \Rightarrow 2x = \begin{bmatrix} 8 & 8 \\ 0 & 8 \end{bmatrix}$$

$$x = \begin{bmatrix} 4 & 4 \\ 0 & 4 \end{bmatrix}$$

$$\textcircled{2} \Rightarrow x - \begin{bmatrix} 3 & 6 \\ 0 & -1 \end{bmatrix} = y$$

$$\begin{bmatrix} 4 & 4 \\ 0 & 4 \end{bmatrix} - \begin{bmatrix} 3 & 6 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 0 & 5 \end{bmatrix} = y$$

*] Find the values of x & y :

$$2 \begin{bmatrix} x & 5 \\ 7 & y-3 \end{bmatrix} + \begin{bmatrix} 3 & -4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 0 & 6 \\ 15 & 14 \end{bmatrix}$$

$$\begin{bmatrix} 2x & 10 \\ 14 & 2y-6 \end{bmatrix} + \begin{bmatrix} 3 & -4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 0 & 6 \\ 15 & 14 \end{bmatrix}$$

$$\begin{bmatrix} 2x & 10 \\ 14 & 2y-6 \end{bmatrix} = \begin{bmatrix} 4 & -4 \\ 14 & 12 \end{bmatrix}$$

$$\begin{aligned} 2x + 3 &= 7 & 2y - 6 + 2 &= 14 \\ 2x &= 4 & 2y &= 18 \\ \boxed{x=2} & & \boxed{y=9} & \end{aligned}$$

Ex Multiplication of matrices :

2 matrices A & B are compatible for multiplication
iff no. of columns in A = no. of rows in B.

$$\text{order (A)} = m \times n$$

$$\text{order (B)} = n \times p$$

$$\text{order (AB)} = m \times p$$

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 4 & 5 \\ 8 & 9 \end{bmatrix} \quad \left. \begin{array}{l} \\ \end{array} \right\} AB \neq BA$$

$$BA = \begin{bmatrix} 0 & -1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} -3 & -4 \\ 11 & 16 \end{bmatrix}$$

$$CD = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 4 & 3 & 5 \\ 4 & 2 & 5 & 7 \end{bmatrix} = \begin{bmatrix} 9 & 8 & 13 & 19 \\ 4 & 2 & 5 & 7 \end{bmatrix}$$

$$DC = \begin{bmatrix} 3 & 2 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 1 & 5 \\ 4 & 1 \end{bmatrix} = \begin{bmatrix} 11 & 17 \\ 22 & 15 \end{bmatrix} \quad \left. \begin{array}{l} \\ \end{array} \right\} CD \neq DC$$

$$DC = \begin{bmatrix} 1 & 5 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 2 & 5 \end{bmatrix} = \begin{bmatrix} 13 & 27 \\ 14 & 13 \end{bmatrix}$$

$$\star] A = \begin{bmatrix} 1 & 2 & 3 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 4 \\ 0 \\ -1 \\ 2 \end{bmatrix}$$

$$AB = [4 + 0 - 3 + 8] = [9]$$

$$BA = \begin{bmatrix} 4 \\ 0 \\ -1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 4 & 8 & 12 & 16 \\ 0 & 0 & 0 & 0 \\ -1 & -2 & -3 & -4 \\ 2 & 4 & 6 & 8 \end{bmatrix}$$

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$$(AB)' = [9]' = [9]$$

$$B'A' = [4 \ 0 \ -1 \ 2] \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = [4+0-3+8] = [9]$$

$$\Rightarrow (AB)' = B'A'$$

*] $A = \begin{bmatrix} 1 & -2 & 3 \\ -4 & 2 & 5 \\ 8 & 11 & -5 \end{bmatrix}_{3 \times 3}$ $B = \begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 2 & 1 \end{bmatrix}_{3 \times 2}$ Show that $AB \neq BA$

$$AB = \begin{bmatrix} 2-8+6 & 3-10+3 \\ -8+8+10 & -12+10+5 \\ 16-11-5 & 22-5-5 \end{bmatrix} = \begin{bmatrix} 0 & -4 \\ 10 & 3 \\ 1 & 12 \end{bmatrix}$$

$$BA = \begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 & 3 \\ -4 & 2 & 5 \end{bmatrix} = \begin{bmatrix} 2-12+9 & -4+6 & 6+15 \\ 4-20 & -8+10 & 12+25 \\ 2-4 & -4+2 & 6+5 \end{bmatrix} = \begin{bmatrix} -10 & 2 & 21 \\ -16 & 2 & 37 \\ -2 & -2 & 11 \end{bmatrix}$$

*] $A = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 0 & 3 \\ 3 & -1 & 2 \end{bmatrix}_{3 \times 3}$ $B = \begin{bmatrix} 1 & -3 \\ 0 & 2 \\ -1 & 4 \end{bmatrix}_{3 \times 2}$ $C = \begin{bmatrix} 1 & 2 & 3 & -4 \\ 2 & 0 & -2 & 1 \end{bmatrix}_{2 \times 4}$

Show that $(AB)C = A(BC)$

Sol] $AB = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 0 & 3 \\ 3 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & -3 \\ 0 & 2 \\ -1 & 4 \end{bmatrix} = \begin{bmatrix} 1+0-1 & 3+2-4 \\ 2+0-3 & 0+0+12 \\ 3+0-2 & 9-2+8 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 18 \\ 1 & 15 \end{bmatrix}$

$$(AB)C = \begin{bmatrix} 0 & 1 \\ -1 & 18 \\ 1 & 15 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & -4 \\ 2 & 0 & -2 & 1 \end{bmatrix} = \begin{bmatrix} 0+2 & 2+0 & 0-2 & 0-4 \\ -1+36 & -2+0 & -3-36 & 4+18 \\ 1+30 & 2+0 & 3-30 & -4+15 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 2 & -2 & -4 \\ 35 & -2 & -39 & 22 \\ 31 & 2 & -27 & 11 \end{bmatrix}$$

Achiever

$$\begin{aligned}
 BC &= \begin{bmatrix} 1 & 3 \\ 0 & 2 \\ -1 & 4 \end{bmatrix}_{3 \times 2} \begin{bmatrix} 1 & 2 & 3 & -4 \\ 2 & 0 & -2 & 1 \end{bmatrix}_{2 \times 4} = \begin{bmatrix} 1+6 & -2+0 & 3-6 & -4+3 \\ 0+4 & 0+0 & 0-4 & 0+2 \\ -1+8 & -2+0 & -3-8 & 4+4 \end{bmatrix} \\
 &= \begin{bmatrix} 7 & 2 & -3 & -1 \\ 4 & 0 & -4 & 2 \\ 7 & -2 & -11 & 8 \end{bmatrix}_{3 \times 4} \\
 A(BC) &= \begin{bmatrix} 1 & 2 & 3 \\ 3 & 0 & 1 \\ 3 & -1 & 2 \end{bmatrix}_{3 \times 3} \begin{bmatrix} 7 & 2 & -3 & -1 \\ 4 & 0 & -4 & 2 \\ 7 & -2 & -11 & 8 \end{bmatrix}_{3 \times 4} = A \quad [*] \\
 &= \begin{bmatrix} 7+4-7 & 2+0+2 & -3-4+11 & -1+2-8 \\ 14+0+21 & 4+0-6 & -6-0-33 & 8-2+0+24 \\ 21-4+14 & 6+0-4 & -9+4-22 & -3-2+16 \end{bmatrix} \\
 &= \begin{bmatrix} 4 & 4 & 4 & -7 \\ 35 & -2 & -39 & 22 \\ 31 & 2 & -21 & 11 \end{bmatrix} \rightarrow (2) \quad \therefore (1) = (2) \Rightarrow (AB)C = A(BC)
 \end{aligned}$$

★] $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & -2 & 1 \\ 4 & 2 & 1 \end{bmatrix}$ Show that $A^3 - 23A - 40I = 0$

Sol] $A^2 = \begin{bmatrix} 1 & 2 & 3 \\ 3 & -2 & 1 \\ 4 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 3 & -2 & 1 \\ 4 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1+6+12 & 2-4+6 & 3+2+3 \\ 3-6+4 & 6+4+2 & 9-2+1 \\ 4+6+4 & 8-4+2 & 12+2+1 \end{bmatrix}$

$$= \begin{bmatrix} 19 & 4 & 8 \\ 1 & 12 & 8 \\ 14 & 6 & 15 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 3 & -2 & 1 \\ 4 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 19+12+32 & 38-8+16 & 57+4+8 \\ 1+36+32 & 2-24+16 & 3+12+8 \\ 14+18+60 & 28-12+30 & 42+6+15 \end{bmatrix}$$

$$= \begin{bmatrix} 63 & 46 & 69 \\ 69 & -6 & 23 \\ 92 & 46 & 63 \end{bmatrix} = A^3 \rightarrow (3)$$

$$23A = 23 \begin{bmatrix} 1 & 2 & 3 \\ 3 & -2 & 1 \\ 4 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 23 & 46 & 69 \\ 69 & -46 & 23 \\ 92 & 46 & 23 \end{bmatrix} \rightarrow \textcircled{2}$$

$$40I = 40 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 40 & 0 & 0 \\ 0 & 40 & 0 \\ 0 & 0 & 40 \end{bmatrix} \rightarrow \textcircled{3}$$

From $\textcircled{1}$ & $\textcircled{2}$, $\textcircled{1} - \textcircled{2} =$

$$\begin{bmatrix} 63 & 46 & 69 \\ 69 & -6 & 23 \\ 92 & 46 & 63 \end{bmatrix} - \begin{bmatrix} 23 & 46 & 69 \\ 69 & -46 & 23 \\ 92 & 46 & 23 \end{bmatrix} = \begin{bmatrix} 40 & 0 & 0 \\ 0 & 40 & 0 \\ 0 & 0 & 40 \end{bmatrix} \rightarrow \textcircled{4}$$

$\textcircled{4} - \textcircled{3} =$

$$\begin{bmatrix} 40 & 0 & 0 \\ 0 & 40 & 0 \\ 0 & 0 & 40 \end{bmatrix} - \begin{bmatrix} 40 & 0 & 0 \\ 0 & 40 & 0 \\ 0 & 0 & 40 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 0$$

*] Construct a 2×2 matrix whose elements are given by $(i) a_{ij} = \frac{(i-j)^2}{2}$

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} 1/2 & 9/2 \\ 0 & 2 \end{bmatrix}$$

$$(ii) a_{ij} = |2i + 3j|$$

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} 1/2 & 2 \\ 1/2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} x^2/20 & x^5/112 \\ x^5/112 & x^2/20 \end{bmatrix} + \begin{bmatrix} x^2/20 & x^5/112 \\ x^5/112 & x^2/20 \end{bmatrix}$$

Achiever

EXERCISE 3.2

$$1] A = \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix}, B = \begin{bmatrix} 1 & 3 \\ -2 & 5 \end{bmatrix}, C = \begin{bmatrix} -2 & 5 \\ 3 & 4 \end{bmatrix}$$

Find: (i) $A+B$ (ii) $A-B$ (iii) $3A-C$ (iv) AB (v) BA

$$(i) \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 3 \\ -2 & 5 \end{bmatrix} = \begin{bmatrix} 3 & 7 \\ 1 & 7 \end{bmatrix}$$

$$(ii) \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix} - \begin{bmatrix} 1 & 3 \\ -2 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 5 & -3 \end{bmatrix}$$

$$(iii) \begin{bmatrix} 6 & 12 \\ 9 & 6 \end{bmatrix} - \begin{bmatrix} -2 & 5 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 8 & 7 \\ 6 & 2 \end{bmatrix}$$

$$(iv) AB = \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ -2 & 5 \end{bmatrix} = \begin{bmatrix} -6 & 26 \\ -1 & 19 \end{bmatrix}$$

$$(v) BA = \begin{bmatrix} 1 & 3 \\ -2 & 5 \end{bmatrix} \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 11 & 10 \\ 11 & 2 \end{bmatrix}$$

$$2] (i) \begin{bmatrix} a & b \\ -b & a \end{bmatrix} + \begin{bmatrix} a & b \\ b & a \end{bmatrix} = \begin{bmatrix} 2a & 2b \\ 0 & 2a \end{bmatrix} = 2 \begin{bmatrix} a & b \\ 0 & a \end{bmatrix}$$

$$(ii) \begin{bmatrix} a^2+b^2 & b^2+c^2 \\ a^2+c^2 & a^2+b^2 \end{bmatrix} + \begin{bmatrix} 2ab & 2bc \\ -2ac & -2ab \end{bmatrix} = \begin{bmatrix} (a+b)^2 & (b+c)^2 \\ (a-c)^2 & (a-b)^2 \end{bmatrix}$$

$$(iii) \begin{bmatrix} -1 & 4 & -6 \\ 8 & 5 & 16 \\ 2 & 8 & 5 \end{bmatrix} + \begin{bmatrix} 12 & 7 & 6 \\ 8 & 0 & 5 \\ 3 & 2 & 4 \end{bmatrix} = \begin{bmatrix} 11 & 11 & 0 \\ 16 & 5 & 21 \\ 5 & 10 & 9 \end{bmatrix}$$

$$(iv) \begin{bmatrix} \cos^2 x & \sin^2 x \\ \sin^2 x & \cos^2 x \end{bmatrix} + \begin{bmatrix} \sin^2 x & \cos^2 x \\ \cos^2 x & \sin^2 x \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

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(3) (i) $\begin{bmatrix} a & b \\ -b & a \end{bmatrix} \begin{bmatrix} a & -b \\ b & a \end{bmatrix} = \begin{bmatrix} a^2+b^2 & 0 \\ 0 & a^2+b^2 \end{bmatrix} = a^2+b^2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = (a^2+b^2)I_2$

(ii) $\begin{bmatrix} 1 & -2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} -3 & -4 & 1 \\ 8 & 13 & 9 \end{bmatrix}$

(iii) $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 2 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 3 & 4 \\ 4 & 6 & 8 \\ 6 & 9 & 12 \end{bmatrix}$

(iv) $\begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & -3 & 5 \\ 0 & 2 & 4 \\ 3 & 0 & 5 \end{bmatrix} = \begin{bmatrix} 14 & 0 & 42 \\ 18 & -1 & 56 \\ 22 & -2 & 70 \end{bmatrix}$

(v) $\begin{bmatrix} 2 & 1 \\ 3 & 2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ -1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 5 \\ -2 & 2 & 0 \end{bmatrix}$

(vi) $\begin{bmatrix} 3 & -1 & 3 \\ -1 & 0 & 2 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 2 & -3 \\ 1 & 0 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 14 & -6 \\ 4 & 5 \end{bmatrix}$

4. $A = \begin{bmatrix} 1 & 2 & -3 \\ 5 & 0 & 2 \\ 1 & -1 & 1 \end{bmatrix}$ $B = \begin{bmatrix} 3 & -1 & 2 \\ 4 & 2 & 5 \\ 2 & 0 & 3 \end{bmatrix}$ $C = \begin{bmatrix} 4 & 12 \\ 0 & 3 & 2 \\ 1 & -2 & 3 \end{bmatrix}$

Verify: $A+(B-C) = (A+B)-C$

Sol] $A+(B-C) = \begin{bmatrix} 1 & 2 & -3 \\ 5 & 0 & 2 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} -1 & -2 & 0 \\ 4 & -1 & 3 \\ 1 & 2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & -3 \\ 9 & -1 & 5 \\ 2 & 1 & 1 \end{bmatrix} \rightarrow \textcircled{1}$

$(A+B)-C = \begin{bmatrix} 4 & 1 & -1 \\ 9 & 2 & 7 \\ 3 & -1 & 4 \end{bmatrix} \begin{bmatrix} 4 & 1 & 2 \\ 0 & 3 & 2 \\ 1 & -2 & 3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & -3 \\ 9 & -1 & 5 \\ 2 & 1 & 1 \end{bmatrix} \rightarrow \textcircled{2}$

$\therefore \textcircled{1} = \textcircled{2}$

$\therefore A+(B-C) = (A+B)-C$

Achiever

14.5 H.W

5] Simplify $\cos \theta \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} + \sin \theta \begin{bmatrix} \sin \theta & -\cos \theta \\ \cos \theta & \sin \theta \end{bmatrix}$

Sol] $\begin{bmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ -\cos \theta \sin \theta & \cos^2 \theta \end{bmatrix} + \begin{bmatrix} \sin^2 \theta & -\cos \theta \sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{bmatrix}$
 $= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

7] (i) $x + y = \begin{bmatrix} 7 & 0 \\ 2 & 5 \end{bmatrix} \rightarrow \textcircled{1} \quad x - y = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \rightarrow \textcircled{2}$

$\textcircled{2} + \textcircled{1} = 2x = \begin{bmatrix} 10 & 0 \\ 2 & 8 \end{bmatrix} \Rightarrow x = \begin{bmatrix} 5 & 0 \\ 1 & 4 \end{bmatrix}$

$\textcircled{1} \Rightarrow y = \begin{bmatrix} 7 & 0 \\ 2 & 5 \end{bmatrix} - x = \begin{bmatrix} 7 & 0 \\ 2 & 5 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$

(ii) $2x + 3y = \begin{bmatrix} 2 & 3 \\ 4 & 0 \end{bmatrix} \rightarrow \textcircled{1} \quad 3x + 2y = \begin{bmatrix} 2 & -2 \\ -1 & 5 \end{bmatrix} \rightarrow \textcircled{2}$

$2x + 3y = \begin{bmatrix} 2 & 3 \\ 4 & 0 \end{bmatrix} \times 3$

$3x + 2y = \begin{bmatrix} 2 & -2 \\ -1 & 5 \end{bmatrix} \times 2$

$\Rightarrow 6x + 9y = \begin{bmatrix} 6 & 9 \\ 12 & 0 \end{bmatrix}$

$6x + 4y = \begin{bmatrix} 4 & -4 \\ -2 & 10 \end{bmatrix}$

$\ominus$
 $5y = \begin{bmatrix} 2 & 13 \\ 14 & -10 \end{bmatrix}$

$y = \begin{bmatrix} 2/5 & 13/5 \\ 14/5 & -2 \end{bmatrix}$

$$\textcircled{1} \Rightarrow 2X = \begin{bmatrix} 2 & 3 \\ 4 & 0 \end{bmatrix} - 3Y$$

$$2X = \begin{bmatrix} 2 & 3 \\ 4 & 0 \end{bmatrix} - \begin{bmatrix} 6/5 & 39/5 \\ 22/5 & -6 \end{bmatrix}$$

$$2X = \begin{bmatrix} 4/5 & -24/5 \\ -22/5 & 6 \end{bmatrix}$$

$$X = \begin{bmatrix} 2/5 & -12/5 \\ -11/5 & 3 \end{bmatrix}$$

$$8] \text{ Find } x, y \text{ if } \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}, 2X + Y = \begin{bmatrix} 1 & 0 \\ -3 & 2 \end{bmatrix}$$

$$\text{Sol]} 2X = \begin{bmatrix} 1 & 0 \\ -3 & 2 \end{bmatrix} - \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & -2 \\ -4 & -2 \end{bmatrix}$$

$$X = \begin{bmatrix} -1 & -1 \\ -2 & -1 \end{bmatrix}$$

$$9] \text{ Find } x, y \text{ if } 2 \begin{bmatrix} 1 & 3 \\ 0 & x \end{bmatrix} + \begin{bmatrix} y & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 6 \\ 1 & 8 \end{bmatrix}$$

$$\text{Sol]} 2x + 2 = 8 \quad 0 + y = 5$$

$$x = \frac{6}{2} = 3 \quad y = 5 - 2 = 3$$

$$\boxed{x=3}$$

$$\boxed{y=3}$$

$$10] \text{ Solve: } 2 \begin{bmatrix} x & 8 \\ y & t \end{bmatrix} + 3 \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} = 3 \begin{bmatrix} 3 & 5 \\ 4 & 6 \end{bmatrix}$$

Achiever

Sol] $2x + 3 = 9$ $2z - 3 = 15$ $2y + 0 = 12$

$2x = 6$ $2z = 18$ $2y = 12$

$\boxed{x = 3}$ $\boxed{z = 9}$ $\boxed{y = 6}$

$2t + 6 = 18$

$2t = 18 - 6$

$\boxed{t = 6}$

11] $x \begin{bmatrix} 2 \\ 3 \end{bmatrix} + y \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \end{bmatrix}$ find x, y .

Sol] $2x - y = 10$

$3x + y = 5$

$5x = 15$

$\boxed{x = 3}$

$6 - 10 = y$

$\boxed{y = -4}$

12] $3 \begin{bmatrix} x & y \\ z & w \end{bmatrix} = \begin{bmatrix} x & 6 \\ -1 & 2w \end{bmatrix} + \begin{bmatrix} 4 & x+y \\ z+w & 3 \end{bmatrix}$ find x, y, z, w .

Sol] $3x = x + 4$ $3y = 6 + x + y$ $3z = -1 + z + w$

$2x = 4$ $2y = 8$ $2z = -1 + w$

$\boxed{x = 2}$

$\boxed{y = 4}$

$2z = 2 \Rightarrow \boxed{z = 1}$

$3w = 2w + 3$

$\boxed{w = 3}$

13] $F(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$ ST $F(x) \cdot F(y) = F(x+y)$

$F(x) \cdot F(y) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos y & -\sin y & 0 \\ \sin y & \cos y & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$= \begin{bmatrix} \cos x \cos y - \sin x \sin y & 0 & \cos x \sin y - \sin x \cos y \\ \sin x \cos y + \cos x \sin y & 0 & -\sin x \sin y + \cos x \cos y \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos(x+y) & 0 & 0 \\ \sin(x+y) & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = F(x+y) = \text{RHS}$$

15] $A = \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix}$ Find $A^2 - 5A + 6I$

Sol] $A^2 = \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 8 & -1 & 2 \\ 9 & -2 & 5 \\ 0 & -1 & -2 \end{bmatrix}$

$$5A = \begin{bmatrix} 10 & 0 & 5 \\ 10 & 5 & 15 \\ 5 & -5 & 0 \end{bmatrix}$$

$$6I = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

$$A^2 - 5A = \begin{bmatrix} -5 & -1 & -3 \\ -1 & -7 & -10 \\ -5 & 4 & -2 \end{bmatrix}$$

$$\begin{bmatrix} -5 & -1 & -3 \\ -1 & -7 & -10 \\ -5 & 4 & -2 \end{bmatrix} + \begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix} = \begin{bmatrix} +1 & -1 & -3 \\ -1 & -1 & -10 \\ -5 & 4 & 4 \end{bmatrix}$$

16] $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$ ST: $A^3 - 6A^2 + 7A + 2I = 0$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = X \quad \text{Achiever}$$

$$\text{Sol)} \quad A^2 = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{bmatrix}$$

$$A^3 = A^2(A) = \begin{bmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 21 & 0 & 34 \\ 12 & 8 & 23 \\ 34 & 0 & 55 \end{bmatrix}$$

$$6A^2 = \begin{bmatrix} 30 & 0 & 48 \\ 12 & 24 & 30 \\ 48 & 0 & 78 \end{bmatrix} \quad 7A = \begin{bmatrix} 7 & 0 & 14 \\ 0 & 14 & 7 \\ 14 & 0 & 21 \end{bmatrix}$$

$$2I = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\begin{aligned} \text{LHS} \quad A^3 - 6A^2 + 7A + 2I &= \begin{bmatrix} 21 & 0 & 34 \\ 12 & 8 & 23 \\ 34 & 0 & 55 \end{bmatrix} - \begin{bmatrix} 30 & 0 & 48 \\ 12 & 24 & 30 \\ 48 & 0 & 78 \end{bmatrix} + \begin{bmatrix} 7 & 0 & 14 \\ 0 & 14 & 7 \\ 14 & 0 & 21 \end{bmatrix} + \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \mathbf{O} = \text{RHS} \end{aligned}$$

$$17) \quad A = \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix} \quad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{find } k \text{ so that } A^2 = kA - 2I$$

$$\text{Sol)} \quad \text{LHS: } A^2 = \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 4 & -4 \end{bmatrix}$$

$$\text{RHS: } kA - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 \\ 4 & -4 \end{bmatrix} = kA - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$kA = \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix}$$

$$k = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Eg 21] $A = \begin{bmatrix} -2 \\ 4 \\ 5 \end{bmatrix}_{3 \times 1}$ $B = \begin{bmatrix} 1 & 3 & -6 \end{bmatrix}_{1 \times 3}$, verify $(AB)' = B'A'$

Sol] $AB = \begin{bmatrix} -2 \\ 4 \\ 5 \end{bmatrix} \begin{bmatrix} 1 & 3 & -6 \end{bmatrix} = \begin{bmatrix} -2 & -6 & 12 \\ 4 & 12 & -24 \\ 5 & 15 & -30 \end{bmatrix}$

$(AB)' = \begin{bmatrix} -2 & 4 & 5 \\ -6 & 12 & 15 \\ 12 & -24 & -30 \end{bmatrix} \rightarrow \textcircled{1}$

$B' = \begin{bmatrix} 1 \\ 3 \\ -6 \end{bmatrix}$ $A' = \begin{bmatrix} -2 & 4 & 5 \end{bmatrix}$

$\therefore B'A' = \begin{bmatrix} 1 \\ 3 \\ -6 \end{bmatrix} \begin{bmatrix} -2 & 4 & 5 \end{bmatrix} = \begin{bmatrix} -2 & 4 & 5 \\ -6 & 12 & 15 \\ 12 & -24 & -30 \end{bmatrix} \rightarrow \textcircled{2}$

$\therefore \textcircled{1} = \textcircled{2}$

$\therefore (AB)' = B'A'$

5] $A = \begin{bmatrix} 2/3 & 1 & 5/3 \\ 1/3 & 2/3 & 4/3 \\ 1/3 & 2 & 2/3 \end{bmatrix}$ $B = \begin{bmatrix} 2/5 & 3/5 & 1 \\ 1/5 & 2/5 & 4/5 \\ 7/5 & 6/5 & 2/5 \end{bmatrix}$ Find: $3A - 5B$

Sol] $3A = \begin{bmatrix} 2 & 3 & 5 \\ 1 & 2 & 4 \\ 1 & 6 & 2 \end{bmatrix}$

$5B = \begin{bmatrix} 2 & 3 & 5 \\ 1 & 2 & 4 \\ 7 & 6 & 2 \end{bmatrix}$

$\therefore 3A - 5B = \begin{bmatrix} 2 & 3 & 5 \\ 1 & 2 & 4 \\ 1 & 6 & 2 \end{bmatrix} - \begin{bmatrix} 2 & 3 & 5 \\ 1 & 2 & 4 \\ 7 & 6 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Achiever

14] (i) Show that $\begin{bmatrix} 5 & -1 \\ 6 & 7 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix} \neq \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & -1 \\ 6 & 7 \end{bmatrix}$

Sol] $\begin{bmatrix} 7 & 1 \\ 33 & 34 \end{bmatrix} \neq \begin{bmatrix} 16 & 5 \\ 39 & 25 \end{bmatrix}$

(ii) $\begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 2 & 3 & 4 \end{bmatrix} \neq \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$

Sol] $\begin{bmatrix} 5 & 8 & 14 \\ 0 & -1 & 1 \\ 0 & -1 & 1 \end{bmatrix} \neq \begin{bmatrix} -1 & -1 & -3 \\ 1 & 0 & 0 \\ 2 & 6 & -11 \end{bmatrix}$

EXERCISE 3.3

1] Find transpose:

(i) $A = \begin{bmatrix} 5 \\ 1/2 \\ -1 \end{bmatrix}_{3 \times 1}$ $A' = \begin{bmatrix} 5 & 1/2 & -1 \end{bmatrix}_{1 \times 3}$

NOTE: (i) If order(A) = $m \times n$ then order(A') = $n \times m$

(ii) $(A')' = A$

(iii) $(KA)' = KA'$

(iv) $(A+B)' = A' + B'$

(v) $(AB)' = B'A'$

** (vi) Every square matrix can be expressed as the sum of a symmetric and a skew symmetric matrix.

(ii) $A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$ $A' = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$

(iii) $A = \begin{bmatrix} -1 & 5 & 6 \\ \sqrt{3} & 5 & 6 \\ 2 & 3 & -1 \end{bmatrix}$ $A' = \begin{bmatrix} -1 & \sqrt{3} & 2 \\ 5 & 5 & 3 \\ 6 & 6 & -1 \end{bmatrix}$

2] $A = \begin{bmatrix} -1 & 2 & 3 \\ 5 & 7 & 9 \\ -2 & 1 & 1 \end{bmatrix}$ $B = \begin{bmatrix} -4 & 1 & -5 \\ 1 & 2 & 0 \\ 1 & 3 & 1 \end{bmatrix}$ Verify that

(i) $(A+B)' = A' + B'$ (ii) $(A-B)' = A' - B'$

Sol.] i] $A+B = \begin{bmatrix} -5 & 3 & -2 \\ 6 & 9 & 9 \\ -1 & 4 & 2 \end{bmatrix}$

$(A+B)' = \begin{bmatrix} -5 & 6 & -1 \\ 3 & 9 & 4 \\ -2 & 9 & 2 \end{bmatrix}$

$A' = \begin{bmatrix} -1 & 5 & -2 \\ 2 & 7 & 1 \\ 3 & 9 & 1 \end{bmatrix}$ $B' = \begin{bmatrix} -4 & 1 & 1 \\ 1 & 2 & 3 \\ -5 & 0 & 1 \end{bmatrix}$

$A' + B' = \begin{bmatrix} -5 & 6 & -1 \\ 3 & 9 & 4 \\ -2 & 9 & 2 \end{bmatrix}$ $\therefore (A+B)' = A' + B'$

ii] $A-B = \begin{bmatrix} 3 & 1 & 8 \\ 4 & 5 & 9 \\ -3 & -2 & 0 \end{bmatrix}$ $(A-B)' = \begin{bmatrix} 3 & 4 & -3 \\ 1 & 5 & -2 \\ 8 & 9 & 0 \end{bmatrix}$

$A' = \begin{bmatrix} -1 & 5 & -2 \\ 2 & 7 & 1 \\ 3 & 9 & 1 \end{bmatrix}$ $B' = \begin{bmatrix} -4 & 1 & 1 \\ 1 & 2 & 3 \\ -5 & 0 & 1 \end{bmatrix}$

$A' - B' = \begin{bmatrix} 3 & 4 & -3 \\ 1 & 5 & -2 \\ 8 & 9 & 0 \end{bmatrix}$ $\therefore (A-B)' = A' - B'$

3] $A' = \begin{bmatrix} 3 & 4 \\ -1 & 2 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 2 & 1 \\ 1 & 2 & 3 \end{bmatrix}$ then verify

(i) $(A+B)' = A' + B'$ (ii) $(A-B)' = A' - B'$

Sol] i) $A = \begin{bmatrix} 3 & -1 & 0 \\ 4 & 2 & 1 \end{bmatrix}$ $B = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 2 & 3 \end{bmatrix}$

$$A+B = \begin{bmatrix} 2 & 1 & 1 \\ 5 & 4 & 4 \end{bmatrix} \quad (A+B)' = \begin{bmatrix} 2 & 5 \\ 1 & 4 \\ 1 & 4 \end{bmatrix}$$

$$B' = \begin{bmatrix} -1 & 1 \\ 2 & 2 \\ 1 & 3 \end{bmatrix} \quad A' = \begin{bmatrix} 3 & 4 \\ -1 & 2 \\ 0 & 1 \end{bmatrix}$$

$$A'+B' = \begin{bmatrix} 2 & 5 \\ 1 & 4 \\ 1 & 4 \end{bmatrix} \quad \therefore (A+B)' = A'+B'$$

ii) $A-B = \begin{bmatrix} 4 & -3 & -1 \\ 3 & 0 & -2 \end{bmatrix} \quad (A-B)' = \begin{bmatrix} 4 & 3 \\ -3 & 0 \\ -1 & -2 \end{bmatrix}$

$$A'-B' = \begin{bmatrix} 4 & 3 \\ -3 & 0 \\ -1 & -2 \end{bmatrix} \quad \therefore (A-B)' = A'-B'$$

5] Prove: $(AB)' = B'A'$

i) $A = \begin{bmatrix} 1 \\ -4 \\ 5 \end{bmatrix}_{3 \times 1}$ $B = \begin{bmatrix} -1 & 2 & 1 \\ 1 & 2 & 3 \end{bmatrix}_{2 \times 3}$

$$AB = \begin{bmatrix} -1 & 2 & 1 \\ 4 & -8 & -4 \\ -5 & 10 & 5 \end{bmatrix} \quad (AB)' = \begin{bmatrix} -1 & 4 & -5 \\ 2 & -8 & 10 \\ 1 & -4 & 5 \end{bmatrix}$$

$$B' = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} \quad A' = \begin{bmatrix} 1 & -4 & 5 \end{bmatrix}$$

$$B'A' = \begin{bmatrix} -1 & 4 & -5 \\ 2 & -8 & 10 \\ 1 & -4 & 5 \end{bmatrix} \quad \therefore (AB)' = B'A'$$

$$\text{ii)} \quad A = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 5 & 7 \end{bmatrix} \quad \times \text{Ans} = \text{Ans} = A$$

$$AB = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 5 & 7 \\ 2 & 10 & 14 \end{bmatrix} \quad (AB)' = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 5 & 10 \\ 0 & 7 & 14 \end{bmatrix}$$

$$B' = \begin{bmatrix} 1 \\ 5 \\ 7 \end{bmatrix} \quad A' = \begin{bmatrix} 0 & 1 & 2 \end{bmatrix}$$

$$B'A' = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 5 & 10 \\ 0 & 7 & 14 \end{bmatrix} \quad \therefore (AB)' = A'B'$$

$$4] \quad A' = \begin{bmatrix} -2 & 3 \\ 1 & 2 \end{bmatrix} \quad B = \begin{bmatrix} -1 & 0 \\ 1 & 2 \end{bmatrix} \quad \text{Find } (A+2B)'$$

$$\text{Sol]} \quad A = \begin{bmatrix} -2 & 1 \\ 3 & 2 \end{bmatrix} \quad 2B = \begin{bmatrix} -2 & 0 \\ 2 & 4 \end{bmatrix}$$

$$A+2B = \begin{bmatrix} -4 & 1 \\ 5 & 6 \end{bmatrix}$$

$$(A+2B)' = \begin{bmatrix} -4 & 5 \\ 1 & 6 \end{bmatrix}$$

$$6] \quad \text{Verify that } A'A = I$$

$$\text{i)} \quad A = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \quad A' = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$$

$$A'A = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} = A$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

ACHIEVER

NOTE: For any square matrix, if $A'A = I \Rightarrow A$ is orthogonal matrix

$$\text{ii)} \quad A = \begin{bmatrix} \sin \alpha & \cos \alpha \\ -\cos \alpha & \sin \alpha \end{bmatrix}$$

$$A'A = \begin{bmatrix} \sin \alpha & -\cos \alpha \\ \cos \alpha & \sin \alpha \end{bmatrix} \begin{bmatrix} \sin \alpha & \cos \alpha \\ -\cos \alpha & \sin \alpha \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

7] (i) Show that $A = \begin{bmatrix} 1 & -1 & 5 \\ -1 & 2 & 1 \\ 5 & 1 & 3 \end{bmatrix}$ is symmetric.

A Symmetric $\Rightarrow A = A'$

$$A' = \begin{bmatrix} 1 & -1 & 5 \\ -1 & 2 & 1 \\ 5 & 1 & 3 \end{bmatrix}$$

$\therefore A = A' \Rightarrow A$ is a symmetric matrix.

(ii) Show that $A = \begin{bmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix}$ is skew symmetric matrix.

$$A = \begin{bmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix}$$

$$A' = \begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{bmatrix} = - \begin{bmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix} = -A$$

8] $A = \begin{bmatrix} 1 & 5 \\ 6 & 7 \end{bmatrix}$ show that

(i) $(A+A')$ is symmetric

(ii) $(A-A')$ is skew symmetric

Sol.] i] $A = \begin{bmatrix} 1 & 5 \\ -6 & 7 \end{bmatrix}$ $A' = \begin{bmatrix} 1 & 6 \\ 5 & 7 \end{bmatrix}$

$$A + A' = \begin{bmatrix} 2 & 11 \\ 11 & 14 \end{bmatrix}$$

Consider $(A + A')' = \begin{bmatrix} 2 & 11 \\ 11 & 14 \end{bmatrix} = A + A'$

$\therefore (A + A')' = A + A'$, $A + A'$ is symmetric.

ii] $A - A' = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

consider $(A - A')' = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = -\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = -(A - A')$

$\therefore (A - A')' = -(A - A')$

$\therefore A - A'$ is skew symmetric.

13m) ** 10] Express the following as the sum of symmetric and skew symmetric matrix.

Given: $B = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$ $B' = \begin{bmatrix} 2 & -1 & 1 \\ -2 & 3 & -2 \\ -4 & 4 & -3 \end{bmatrix}$

$$P = \frac{1}{2}(B + B') = \frac{1}{2} \begin{bmatrix} 4 & -3 & -3 \\ -3 & 6 & 2 \\ -3 & 2 & -6 \end{bmatrix} = \begin{bmatrix} 2 & -3/2 & -3/2 \\ -3/2 & 3 & 1 \\ -3/2 & 1 & -3 \end{bmatrix}$$

$$P' = \begin{bmatrix} 2 & -3/2 & -3/2 \\ -3/2 & 3 & 1 \\ -3/2 & 1 & -3 \end{bmatrix} = P \Rightarrow P \text{ is symmetric}$$

$$Q = \frac{1}{2}(B - B') = \frac{1}{2} \begin{bmatrix} 0 & -1 & -5 \\ 1 & 0 & 6 \\ 5 & -6 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1/2 & -5/2 \\ 1/2 & 0 & 3 \\ 5/2 & -3 & 0 \end{bmatrix}$$

Achiever

$$Q' = \begin{bmatrix} 0 & \frac{1}{2} & \frac{5}{2} \\ -\frac{1}{2} & 0 & -3 \\ -\frac{5}{2} & 3 & 0 \end{bmatrix} = - \begin{bmatrix} 0 & -\frac{1}{2} & -\frac{5}{2} \\ \frac{1}{2} & 0 & 3 \\ \frac{5}{2} & -3 & 0 \end{bmatrix} = -Q$$

$Q' = -Q \Rightarrow Q$ is skew symmetric

Consider:

$$P + Q = \begin{bmatrix} 2 & -\frac{3}{2} & -\frac{3}{2} \\ -\frac{3}{2} & 3 & 1 \\ -\frac{3}{2} & 1 & -3 \end{bmatrix} + \begin{bmatrix} 0 & -\frac{1}{2} & -\frac{5}{2} \\ \frac{1}{2} & 0 & 3 \\ \frac{5}{2} & -3 & 0 \end{bmatrix} = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} = B$$

$\therefore$ Given matrix has been expressed as sum of a symmetric and a skew symmetric matrix.

$$i) \begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix} = A \quad A' = \begin{bmatrix} 3 & 1 \\ 5 & -1 \end{bmatrix}$$

$$P = \frac{1}{2}(A + A') = \frac{1}{2} \begin{bmatrix} 6 & 6 \\ 6 & -2 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & -1 \end{bmatrix}$$

$$P' = \begin{bmatrix} 3 & 3 \\ 3 & -1 \end{bmatrix} = P \Rightarrow P \text{ is symmetric}$$

$$Q = \frac{1}{2}(A - A') = \frac{1}{2} \begin{bmatrix} 0 & 4 \\ -4 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$$

$$Q' = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix} = - \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix} = -Q$$

$Q' = -Q \Rightarrow Q$ is skew symmetric

Consider:

$$P + Q = \begin{bmatrix} 3 & 3 \\ 3 & -1 \end{bmatrix} + \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix} = A$$

$\therefore$ Given matrix has been expressed as sum of a symmetric and a skew symmetric matrix.

$$\text{iv)] } \begin{bmatrix} 1 & 5 \\ -1 & 2 \end{bmatrix} = A \quad A' = \begin{bmatrix} 1 & -1 \\ 5 & 2 \end{bmatrix}$$

$$P = \frac{1}{2}(A + A') = \frac{1}{2} \begin{bmatrix} 2 & 4 \\ 4 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix}$$

$$P' = \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix} = P \Rightarrow P \text{ is symmetric}$$

$$Q = \frac{1}{2}(A - A') = \frac{1}{2} \begin{bmatrix} 0 & 6 \\ -6 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 3 \\ -3 & 0 \end{bmatrix}$$

$$Q' = \begin{bmatrix} 0 & -3 \\ 3 & 0 \end{bmatrix} = - \begin{bmatrix} 0 & 3 \\ -3 & 0 \end{bmatrix} = -Q$$

$$Q' = -Q \Rightarrow Q \text{ is skew symmetric}$$

Consider

$$P + Q = \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix} + \begin{bmatrix} 0 & 3 \\ -3 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 5 \\ -1 & 2 \end{bmatrix}$$

$$\text{iii)] } A = \begin{bmatrix} 3 & -3 & -1 \\ -2 & -2 & 1 \\ -4 & -5 & 2 \end{bmatrix} \quad A' = \begin{bmatrix} 3 & -2 & -4 \\ -3 & -2 & -5 \\ -1 & 1 & 2 \end{bmatrix}$$

$$P = \frac{1}{2}(A + A') = \frac{1}{2} \begin{bmatrix} 6 & -1 & -5 \\ 1 & -4 & -4 \\ -5 & -4 & 4 \end{bmatrix} = \begin{bmatrix} 3 & -\frac{1}{2} & -\frac{5}{2} \\ \frac{1}{2} & -2 & -2 \\ -\frac{5}{2} & -2 & 2 \end{bmatrix}$$

$$P' = \begin{bmatrix} 3 & -\frac{1}{2} & -\frac{5}{2} \\ \frac{1}{2} & -2 & -2 \\ -\frac{5}{2} & -2 & 2 \end{bmatrix} = P \Rightarrow P \text{ is symmetric}$$

$$Q = \frac{1}{2}(A - A') = \frac{1}{2} \begin{bmatrix} 0 & 5 & 3 \\ -5 & 0 & 6 \\ -3 & -6 & 0 \end{bmatrix} = \begin{bmatrix} 0 & \frac{5}{2} & \frac{3}{2} \\ -\frac{5}{2} & 0 & 3 \\ -\frac{3}{2} & -3 & 0 \end{bmatrix}$$

$$Q' = \begin{bmatrix} 0 & -\frac{5}{2} & -\frac{3}{2} \\ \frac{5}{2} & 0 & -3 \\ \frac{3}{2} & 3 & 0 \end{bmatrix} = - \begin{bmatrix} 0 & \frac{5}{2} & \frac{3}{2} \\ -\frac{5}{2} & 0 & 3 \\ -\frac{3}{2} & -3 & 0 \end{bmatrix} = -Q$$

Achiever

$$Q' = -Q \Rightarrow Q \text{ is skew symmetric}$$

Consider

$$P + Q = \begin{bmatrix} 3 & \frac{1}{2} & -\frac{5}{2} \\ \frac{1}{2} & -2 & -2 \\ -\frac{5}{2} & -2 & 2 \end{bmatrix} + \begin{bmatrix} 0 & \frac{5}{2} & \frac{3}{2} \\ -\frac{5}{2} & 0 & 3 \\ -\frac{3}{2} & -3 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 3 & -1 \\ -2 & -2 & 1 \\ -4 & -5 & 2 \end{bmatrix}$$

$$\text{ii)} \quad \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix} = A \quad A' = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

$$P = \frac{1}{2}(A + A') = \frac{1}{2} \begin{bmatrix} 12 & -4 & 4 \\ -4 & 6 & -2 \\ 4 & -2 & 6 \end{bmatrix} = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

$$P' = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix} = P \Rightarrow P \text{ is symmetric}$$

$$Q = \frac{1}{2}(A - A') = \frac{1}{2} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$Q' = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = - \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = -Q$$

$$Q' = -Q \Rightarrow Q \text{ is skew symmetric}$$

Consider,

$$P + Q = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix} = A$$

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9] Find $\frac{1}{2}(A+A')$ and $\frac{1}{2}(A-A')$, when

$$A = \begin{bmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{bmatrix}$$

Sol] $A = \begin{bmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{bmatrix}$ $A' = \begin{bmatrix} 0 & -a & -b \\ a & 0 & -c \\ b & c & 0 \end{bmatrix}$

$$\frac{1}{2}(A+A') = \frac{1}{2} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$
$$\frac{1}{2}(A-A') = \frac{1}{2} \begin{bmatrix} 0 & 2a & 2b \\ -2a & 0 & 2c \\ -2b & -2c & 0 \end{bmatrix} = \begin{bmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{bmatrix}$$

(3m) **

Invertible Matrices

If A is a square matrix of order n and if there exists another square matrix B of same order n , such that $AB = BA = I$, then B is called the inverse matrix of A and is denoted by A^{-1} .

Properties under matrix multiplication:

Associative property: matrix X^n is always associative
 $A(BC) = (AB)C$

Distributive property: $A(B+C) = AB + AC$

** Commutative property: In general, matrix X^n is not commutative i.e., $AB \neq BA$

Existence of Identity:

Unit/Identity matrix ' I ' is multiplicative identity in matrices

$$\therefore AI = IA = A \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = A$$

Existence of Inverse:

(a) If $AB = I$ then $A^{-1} = B$ and $B^{-1} = A$

(b) If A is singular matrix then $A^{-1} = \frac{1}{|A|} (\text{adj } A)$

If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then

(a) $|A| = ad - bc$

(b) $\text{adj } A = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

(c) $A^{-1} = \frac{1}{|A|} \text{adj } A = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

ACHIEVER

Find the inverse of the following:

$$(1) A = \begin{bmatrix} 2 & 3 \\ -5 & 1 \end{bmatrix}$$

$$|A| = 2 - (-15) = 17$$

$$\text{adj } A = \begin{bmatrix} 1 & -3 \\ 5 & 2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{adj } A = \frac{1}{17} \begin{bmatrix} 1 & -3 \\ 5 & 2 \end{bmatrix}$$

$$(2) A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$|A| = 4 - 6 = -2$$

$$\text{adj } A = \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{adj } A = \frac{1}{-2} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$$

$$(3) A = \begin{bmatrix} 1 & 6 \\ 2 & 12 \end{bmatrix}$$

$$|A| = 12 - 12 = 0$$

$\Rightarrow A$ is singular matrix

$\therefore A^{-1}$ does not exist

$$(4) A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

$$|A| = 4 - 4 = 0$$

A^{-1} does not exist

$$(5) A = \begin{bmatrix} 5 & -6 \\ 4 & 3 \end{bmatrix}$$

$$|A| = 15 - (-24) = 39$$

$$A^{-1} = \frac{1}{|A|} \text{adj } A = \frac{1}{39} \begin{bmatrix} 3 & 6 \\ -4 & 5 \end{bmatrix}$$

$$(6) A = \begin{bmatrix} -1 & 3 \\ 5 & -7 \end{bmatrix}$$

$$|A| = 7 - 15 = -8$$

$$A^{-1} = \frac{1}{|A|} \text{adj } A = \frac{1}{-8} \begin{bmatrix} -7 & -3 \\ -5 & -1 \end{bmatrix}$$

$$= \frac{1}{8} \begin{bmatrix} 7 & 3 \\ 5 & 1 \end{bmatrix}$$

$$(7) \quad A = \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix} \quad |A| = 0 - (-2) = 2$$

$$A^{-1} = \frac{1}{|A|} \text{adj} A = \frac{1}{2} \begin{bmatrix} 0 & -2 \\ 1 & 1 \end{bmatrix}$$

$$(8) \quad A = \begin{bmatrix} 4 & 3 \\ 2 & 7 \end{bmatrix} \quad |A| = 28 - 6 = 22$$

$$A^{-1} = \frac{1}{22} \begin{bmatrix} 7 & -3 \\ -2 & 4 \end{bmatrix}$$

$$(9) \quad A = \begin{bmatrix} 10 & 12 \\ -14 & -15 \end{bmatrix} \quad |A| = -150 - (-168) = 18$$

$$A^{-1} = \frac{1}{18} \begin{bmatrix} -15 & -12 \\ 14 & 10 \end{bmatrix}$$

$$(10) \quad A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad |A| = 1 - 0 = 1$$

$$A^{-1} = 1 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

NOTE: Properties of ' I '

$$(a) \quad I^n = I$$

$$(b) \quad I^{-1} = I$$

$$(11) \quad A = \begin{bmatrix} 1 & -1 \\ 2 & -4 \end{bmatrix} \quad |A| = -4 - (-2) = -2$$

$$A^{-1} = \frac{1}{-2} \begin{bmatrix} -4 & 1 \\ -2 & 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 4 & -1 \\ 2 & -1 \end{bmatrix}$$

Find the inverses of the following by elementary operation

NOTE: (a) Interchange of 2 rows or 2 columns is allowed

$$R_i \leftrightarrow R_j / C_i \leftrightarrow C_j$$

(b) Scalar $\times$ or scalar division of any row or column is allowed i.e, kR_i or $\frac{R_i}{k} / kC_i$ or $\frac{C_i}{k}$

(c) A multiple of one row can be added or subtracted to another row and vice versa.

$$\text{i.e., } R_i \rightarrow R_i + kR_j / C_i \rightarrow C_i - kC_j$$

(d) To get inverse, we either follow only row transformations or only column transformations.

(e) If row transformation is followed, consider

$$(i) \quad A = I A$$

(ii) Use row transformations to convert A in LHS

to I , and in the process we get $I = BA^{-1}$
Hence, $A^{-1} = B$.

$\downarrow$

$A = IA$

$\begin{bmatrix} 1 & b \\ c & d \end{bmatrix}$

$\downarrow$

$\begin{bmatrix} 1 & b \\ 0 & d \end{bmatrix}$

$\downarrow$

$\begin{bmatrix} 1 & b \\ 0 & 1 \end{bmatrix}$

$\downarrow$

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = BA^{-1}$

(f) If column transformation, consider I

$A = AI$

$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$\downarrow$

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\downarrow$

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$I = AB$
 $A^{-1} = B$

EXERCISE 3.4

Find the inverses of each matrices :

1) $A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$

$A = IA$

$\begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} A$

$R_2 \rightarrow R_2 - 2R_1$

$\begin{bmatrix} 1 & -1 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} A$

$\div R_2 \text{ by } 5$

$\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -\frac{2}{5} & \frac{1}{5} \end{bmatrix} A$

$R_1 \rightarrow R_1 + R_2$

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{3}{5} & \frac{1}{5} \\ -\frac{2}{5} & \frac{1}{5} \end{bmatrix} A$

$I = BA$

$\therefore A^{-1} = B = \begin{bmatrix} \frac{3}{5} & \frac{1}{5} \\ -\frac{2}{5} & \frac{1}{5} \end{bmatrix}$

8) $A = \begin{bmatrix} 4 & 5 \\ 3 & 4 \end{bmatrix}$

$2-A = I A^{-1} A$

$\begin{bmatrix} 4 & 5 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} A$

$R_1 \rightarrow R_1 - R_2$

$\begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} A$

Achiever

$R_2 \rightarrow R_2 - 3R_1$
 $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -3 & 4 \end{bmatrix} A$
 $R_1 \rightarrow R_1 - R_2$
 $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 4 & -5 \\ -3 & 4 \end{bmatrix} A$
 $I = BA$
 $\Rightarrow A^{-1} = B = \begin{bmatrix} 4 & -5 \\ -3 & 4 \end{bmatrix}$

6] $A = \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix}$
 $A = I A$
 $\begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} A$
 $R_1 \rightarrow R_1 - R_2$
 $\begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} A$
 $R_2 \rightarrow R_2 - R_1$
 $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} A$
 $R_1 \rightarrow R_1 - 2R_2$
 $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -5 \\ -1 & 2 \end{bmatrix} A$

2] $A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$
 $A = I A$
 $\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} A$

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$$R_1 \rightarrow R_1 - R_2$$

$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} A$$

$$R_2 \rightarrow R_2 - R_1$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} A$$

$$I = BA$$

$$A^{-1} = B = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$$

3] $A = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}$

$$A = IA$$

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} A$$

$$R_2 \rightarrow R_2 - 2R_1$$

$$\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} A$$

$$R_1 \rightarrow R_1 - 3R_2$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 7 & -3 \\ -2 & 1 \end{bmatrix} A$$

$$I = BA$$

$$A^{-1} = B = \begin{bmatrix} 7 & -3 \\ -2 & 1 \end{bmatrix}$$

** 4] $A = \begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix}$

$$A^{-1} = \frac{1}{|A|} \text{adj } A = -\frac{1}{1} \begin{bmatrix} 7 & -3 \\ -5 & 2 \end{bmatrix} = \begin{bmatrix} -7 & 3 \\ 5 & -2 \end{bmatrix}$$

$$A = A I$$

$$\begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix} = A \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Achiever

$R_2 \leftrightarrow R_1, C_2 \rightarrow C_2 - C_1$
 $\begin{bmatrix} 2 & 1 \\ 5 & 2 \end{bmatrix} = A \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$

$C_1 \leftrightarrow C_2$
 $\begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix} = A \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix}$

$C_2 \rightarrow C_2 - 2C_1$
 $\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} = A \begin{bmatrix} -1 & 3 \\ 1 & -2 \end{bmatrix}$

$C_1 \rightarrow C_1 - 2C_2$
 $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = A \begin{bmatrix} -7 & 3 \\ 5 & -2 \end{bmatrix}$

$I = AB$
 $A^{-1} = B = \begin{bmatrix} -7 & 3 \\ 5 & -2 \end{bmatrix}$

5] $A = \begin{bmatrix} 2 & 1 \\ 7 & 4 \end{bmatrix}$
 $A = AI$
 $\begin{bmatrix} 2 & 1 \\ 7 & 4 \end{bmatrix} = A \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$C_1 \rightarrow C_1 - C_2$
 $\begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix} = A \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$

$C_2 \rightarrow C_2 - C_1$
 $\begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} = A \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$

$C_1 \rightarrow C_1 - 3C_2$
 $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = A \begin{bmatrix} 4 & -1 \\ -7 & 2 \end{bmatrix}$

$I = AB$
 $A^{-1} = B = \begin{bmatrix} 4 & -1 \\ -7 & 2 \end{bmatrix}$

$$7) A = \begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix}$$

$$A = A I$$

$$\begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix} = A \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$C_1 \leftrightarrow C_2$$

$$\begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix} = A \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$C_2 \rightarrow C_2 - 3C_1$$

$$\begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix} = A \begin{bmatrix} 0 & 1 \\ 1 & -3 \end{bmatrix}$$

$$C_2 \rightarrow C_2 / -1$$

$$\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} = A \begin{bmatrix} 0 & -1 \\ 1 & 3 \end{bmatrix}$$

$$C_1 \rightarrow C_1 - 2C_2$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = A \begin{bmatrix} 2 & -1 \\ -5 & 3 \end{bmatrix}$$

$$I = AB$$

$$A^{-1} = B = \begin{bmatrix} 2 & -1 \\ -5 & 3 \end{bmatrix}$$

$$9) A = \begin{bmatrix} 3 & 10 \\ 2 & 7 \end{bmatrix}$$

$$A = I A$$

$$\begin{bmatrix} 3 & 10 \\ 2 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} A$$

$$R_1 \rightarrow R_1 - R_2$$

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} A$$

$$A^{-1} = \frac{1}{|A|} \text{adj} A = \frac{1}{1} \begin{bmatrix} 2 & -1 \\ -5 & 3 \end{bmatrix}$$

$$A \begin{bmatrix} 2 & -1 \\ -5 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\text{or } I = A^{-1} A$$

$$\begin{bmatrix} 1 & -2 \\ 5 & -3 \end{bmatrix} = A^{-1} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$I A^{-1} = A^{-1}$$

$$A \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 10 \\ 2 & 7 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} A = \begin{bmatrix} 3 & 10 \\ 2 & 7 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 5 & -1 \end{bmatrix} A = \begin{bmatrix} 0 & -1 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{2} & 0 \\ \frac{5}{2} & -1 \end{bmatrix} A = \begin{bmatrix} 0 & 1 \\ 1 & -5 \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{2} & 1 \\ \frac{5}{2} & 5 \end{bmatrix} A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{2} & 1 \\ \frac{5}{2} & 5 \end{bmatrix} A = I \quad \text{Achiever}$$

$R_2 \rightarrow R_2 - 2R_1$
 $\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} A$
 $R_1 \rightarrow R_1 - 3R_2$
 $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 7 & -10 \\ -2 & 3 \end{bmatrix} A$
 $I = BA$
 $A^{-1} = B = \begin{bmatrix} 7 & -10 \\ -2 & 3 \end{bmatrix}$

10) $A = \begin{bmatrix} 3 & -1 \\ -4 & 2 \end{bmatrix}$
 $A^{-1} = \frac{1}{20} \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$
 $A = A I$
 $\begin{bmatrix} 3 & -1 \\ -4 & 2 \end{bmatrix} = A \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
 $C_1 \leftrightarrow C_2$
 $\begin{bmatrix} -1 & 3 \\ 2 & -4 \end{bmatrix} = A \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
 $C_1 \rightarrow (-1)C_1$
 $\begin{bmatrix} 1 & 3 \\ -2 & -4 \end{bmatrix} = A \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$
 $C_2 \rightarrow C_2 - 3C_1$
 $\begin{bmatrix} 1 & 0 \\ -2 & -4 \end{bmatrix} = A \begin{bmatrix} 0 & 1 \\ -1 & 3 \end{bmatrix}$
 $C_2 \rightarrow C_2 / 2$
 $\begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} = A \begin{bmatrix} 0 & \frac{1}{2} \\ -1 & \frac{3}{2} \end{bmatrix}$
 $C_1 \rightarrow C_1 + 2C_2$
 $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = A \begin{bmatrix} 1 & \frac{1}{2} \\ 2 & \frac{3}{2} \end{bmatrix}$
 $I = AB$
 $A^{-1} = B = \begin{bmatrix} 1 & \frac{1}{2} \\ 2 & \frac{3}{2} \end{bmatrix}$

$$11] \quad A = \begin{bmatrix} 2 & -6 \\ 1 & -2 \end{bmatrix} \Rightarrow A^{-1} = \frac{1}{2} \begin{bmatrix} -2 & 6 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} -1 & 3 \\ -\frac{1}{2} & 1 \end{bmatrix}$$

$$A = IA$$

$$\begin{bmatrix} 2 & -6 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} A$$

$$R_1 \rightarrow R_1 - R_2$$

$$\begin{bmatrix} 1 & -4 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} A$$

$$R_2 \rightarrow R_2 - R_1$$

$$\begin{bmatrix} 1 & -4 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} A$$

$$R_2 \rightarrow R_2/2$$

$$\begin{bmatrix} 1 & -4 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -\frac{1}{2} & 1 \end{bmatrix} A$$

$$R_1 \rightarrow R_1 + 4R_2$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 3 \\ -\frac{1}{2} & 1 \end{bmatrix} A$$

$$I = BA$$

$$\therefore A^{-1} = B = \begin{bmatrix} -1 & 3 \\ -\frac{1}{2} & 1 \end{bmatrix}$$

$$12] \quad A = \begin{bmatrix} 6 & -3 \\ -2 & 1 \end{bmatrix}$$

$$|A| = 6 - 6 = 0$$

$\Rightarrow A$ is a singular matrix

$\therefore$ Inverse of A does not exist

$$13] \quad A = \begin{bmatrix} 2 & -3 \\ -1 & 2 \end{bmatrix} \quad \frac{1}{5} A^{-1} = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} \quad \frac{1}{5} A^{-1} = A \quad [11]$$

$$A = JA$$

$$R_1 \rightarrow R_1 + R_2$$

$$\begin{bmatrix} 2 & -3 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} A$$

$$R_1 \rightarrow R_1 + R_2$$

$$\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} A$$

$$R_2 \rightarrow R_2 + R_1$$

$$\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} A$$

$$R_1 \rightarrow R_1 + R_2$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} A$$

$$I = BA$$

$$\therefore A^{-1} = B = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$$

$$14] \quad A = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$$

$$|A| = 4 - 4 = 0$$

$\Rightarrow A$ is a singular matrix

$\therefore$ Inverse of matrix A does not exist.

Reason: $\Delta = 0$ for A is a singular matrix.

$$15] \begin{bmatrix} 2 & -3 & 3 \\ 2 & 2 & 3 \\ 3 & -2 & 2 \end{bmatrix} = A$$

$$A = I A$$

$$\begin{bmatrix} 2 & -3 & 3 \\ 2 & 2 & 3 \\ 3 & -2 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$$R_3 \rightarrow R_3 - R_1$$

$$\begin{bmatrix} 2 & -3 & 3 \\ 2 & 2 & 3 \\ 1 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} A$$

$$R_1 \leftrightarrow R_3$$

$$\begin{bmatrix} 1 & 1 & -1 \\ 2 & 2 & 3 \\ 2 & -3 & 3 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} A$$

$$R_2 \rightarrow R_2 - 2R_1; R_3 \rightarrow R_3 - 2R_1$$

$$\begin{bmatrix} 1 & 1 & -1 \\ 0 & 0 & 5 \\ 0 & -5 & 5 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 1 \\ 2 & 1 & -2 \\ 3 & 0 & -2 \end{bmatrix} A$$

$$R_3 \rightarrow R_3 / -5$$

$$\begin{bmatrix} 1 & 1 & -1 \\ 0 & 0 & 5 \\ 0 & 1 & -1 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 1 \\ 2 & 1 & -2 \\ -\frac{3}{5} & 0 & \frac{2}{5} \end{bmatrix} A$$

$$R_2 \leftrightarrow R_3$$

$$\begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 5 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 1 \\ -\frac{3}{5} & 0 & \frac{2}{5} \\ 2 & 1 & -2 \end{bmatrix} A$$

$$R_1 \rightarrow R_1 - R_2$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 5 \end{bmatrix} = \begin{bmatrix} -\frac{2}{5} & 0 & \frac{3}{5} \\ -\frac{3}{5} & 0 & \frac{2}{5} \\ 2 & 1 & -2 \end{bmatrix} A$$

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$$R_3 \rightarrow R_3/5$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -\frac{2}{5} & 0 & \frac{3}{5} \\ -\frac{3}{5} & 0 & \frac{2}{5} \\ \frac{2}{5} & \frac{1}{5} & -\frac{2}{5} \end{bmatrix} A = \begin{bmatrix} \varepsilon & \varepsilon & \varepsilon \\ \varepsilon & \varepsilon & \varepsilon \\ \varepsilon & \varepsilon & \varepsilon \end{bmatrix} \quad [21]$$

$$R_2 \rightarrow R_2 + R_3$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -\frac{2}{5} & 0 & \frac{3}{5} \\ -\frac{1}{5} & \frac{1}{5} & 0 \\ \frac{2}{5} & \frac{1}{5} & -\frac{2}{5} \end{bmatrix} A = \begin{bmatrix} \varepsilon & \varepsilon & \varepsilon \\ \varepsilon & \varepsilon & \varepsilon \\ \varepsilon & \varepsilon & \varepsilon \end{bmatrix}$$

$$I = BA$$

$$A^{-1} = B = \begin{bmatrix} -\frac{2}{5} & 0 & \frac{3}{5} \\ -\frac{1}{5} & \frac{1}{5} & 0 \\ \frac{2}{5} & \frac{1}{5} & -\frac{2}{5} \end{bmatrix}$$

17] $A = \begin{bmatrix} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix}$

$$A = A I$$

$$\begin{bmatrix} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix} = A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$C_1 \rightarrow C_1 + C_3$$

$$\begin{bmatrix} 1 & 0 & -1 \\ 5 & 1 & 0 \\ +3 & 1 & 3 \end{bmatrix} = A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

$$C_3 \rightarrow C_3 + C_1$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 5 & 1 & 5 \\ 3 & 1 & 6 \end{bmatrix} = A \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ +1 & 0 & 2 \end{bmatrix}$$

$$C_1 \rightarrow C_1 - C_3$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 5 \\ -3 & 1 & 6 \end{bmatrix} = A \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -3 & 0 & 2 \end{bmatrix}$$

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$$C_3 \rightarrow C_3 - 5C_2$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 1 & 1 \end{bmatrix} = A \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -5 \\ -1 & 0 & 2 \end{bmatrix}$$

$$C_2 \rightarrow C_2 - C_3; C_1 \rightarrow C_1 + 3C_3$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = A \begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix}$$

$$I = AB$$

$$A^{-1} = B = \begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix}$$

16] $A = \begin{bmatrix} 1 & 3 & -2 \\ -3 & 0 & -5 \\ 2 & 5 & 0 \end{bmatrix}$

$$A = IA$$

$$\begin{bmatrix} 1 & 3 & -2 \\ -3 & 0 & -5 \\ 2 & 5 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = A$$

$$R_2 \rightarrow R_2 + 3R_1$$

$$\begin{bmatrix} 1 & 3 & -2 \\ 0 & 9 & -11 \\ 2 & 5 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = A$$

$$R_3 \rightarrow R_3 - 2R_1$$

$$\begin{bmatrix} 1 & 3 & -2 \\ 0 & 9 & -11 \\ 0 & -1 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} = A$$

Achiever

$R_2 \rightarrow R_2/9$

$$\begin{bmatrix} 1 & 3 & -2 \\ 0 & 1 & -\frac{11}{9} \\ 0 & -1 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{3} & \frac{1}{9} & 0 \\ -2 & 0 & 1 \end{bmatrix} A$$

$R_1 \rightarrow R_1 - 3R_2$

$$\begin{bmatrix} 1 & 0 & \frac{15}{9} \\ 0 & 1 & -\frac{11}{9} \\ 0 & -1 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & \frac{5}{3} \\ 0 & 1 & -\frac{11}{9} \\ -2 & 0 & 1 \end{bmatrix} A$$

$R_3 \rightarrow R_3 + R_2$

$$\begin{bmatrix} 1 & 0 & \frac{15}{9} \\ 0 & 1 & -\frac{11}{9} \\ 0 & 0 & \frac{25}{9} \end{bmatrix} = \begin{bmatrix} 1 & 0 & \frac{5}{3} \\ 0 & 1 & -\frac{11}{9} \\ -\frac{5}{3} & \frac{1}{9} & 1 \end{bmatrix} A$$

$R_3 \rightarrow R_3 \left(\frac{9}{25} \right)$

$$\begin{bmatrix} 1 & 0 & \frac{15}{9} \\ 0 & 1 & -\frac{11}{9} \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & \frac{5}{3} \\ 0 & 1 & -\frac{11}{9} \\ -\frac{3}{5} & \frac{1}{25} & \frac{9}{25} \end{bmatrix} A$$

$R_1 \rightarrow R_1 - \left(\frac{15}{9} \right) R_3$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -\frac{11}{9} \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -\frac{2}{5} \\ 0 & 1 & -\frac{11}{9} \\ -\frac{3}{5} & \frac{1}{25} & \frac{9}{25} \end{bmatrix} A$$

$R_2 \rightarrow R_2 + \left(\frac{11}{9} \right) R_3$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -\frac{2}{5} \\ 0 & 1 & \frac{4}{25} \\ -\frac{3}{5} & \frac{1}{25} & \frac{9}{25} \end{bmatrix} A$$

$I = BA$
 $\therefore A^{-1} = B = \begin{bmatrix} 1 & 0 & -\frac{2}{5} \\ 0 & 1 & \frac{4}{25} \\ -\frac{3}{5} & \frac{1}{25} & \frac{9}{25} \end{bmatrix}$

Ex 2.8] $A = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}$ $B = \begin{bmatrix} 5 & 2 \\ 7 & 4 \end{bmatrix}$ $C = \begin{bmatrix} 2 & 5 \\ 3 & 8 \end{bmatrix}$

Find matrix D such that $CD - AB = O$

Sol] D is of order $2 \times 2 = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$$\therefore \begin{bmatrix} 2 & 5 \\ 3 & 8 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} - \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & 2 \\ 7 & 4 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 2a+5c & 2b+5d \\ 3a+8c & 3b+8d \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 43 & 22 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 2a+5c-3 & 2b+5d-0 \\ 3a+8c-43 & 3b+8d-22 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\therefore 2a+5c-3 = 0 \rightarrow \textcircled{1}$$

$$2b+5d = 0 \rightarrow \textcircled{2}$$

$$3a+8c-43 = 0 \rightarrow \textcircled{3}$$

$$3b+8d-22 = 0 \rightarrow \textcircled{4}$$

Consider $\textcircled{3} - \textcircled{1}$

$$3a+8c = 43$$

$$\ominus 2a+5c = 3$$

$$a+3c = 40$$

$$\Rightarrow a = 40 - 3c \rightarrow \textcircled{5}$$

Substitute $\textcircled{5}$ in $\textcircled{1}$

$$2(40-3c)+5c = 3$$

$$80-6c+5c = 3$$

$$80-3 = c$$

$$\boxed{c = 77} \rightarrow \textcircled{6}$$

Substitute $\textcircled{6}$ in $\textcircled{5}$

$$a = 40 - 3c$$

$$a = 40 - 3(77)$$

$$a = 40 - 231$$

$$\boxed{a = -191}$$

Consider $(4) - (2)$

$$3b + 8d = 22$$

$$2b + 5d = 0$$

$$b + 3d = 22$$

$$\Rightarrow b = 22 - 3d \rightarrow (7)$$

Substitute (7) in (2)

$$2b + 5d = 0$$

$$2(22 - 3d) + 5d = 0$$

$$44 - 6d + 5d = 0$$

$$44 - d = 0$$

$$d = 44 \rightarrow (8)$$

Substitute (8) in (7)

$$b = 22 - 3d$$

$$b = 22 - 3(44)$$

$$b = 22 - 132$$

$$b = -110$$

$$\therefore D = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} -191 & -110 \\ 77 & 44 \end{bmatrix}$$

MISCELLANEOUS EXERCISE

7] For what values of x : $\begin{bmatrix} 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & 1 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ x \end{bmatrix} = 0$?

LHS =

Sol] $\begin{bmatrix} 6 & 2 & 4 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ x \end{bmatrix} = 4 + 4x$

$$4 + 4x = 0$$

$$4x = -4$$

$$x = -1$$

8] $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$, show that $A^2 - 5A + 7I = 0$

Sol] $A^2 = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 8 & 5 \\ -5 & 3 \end{bmatrix}$

$5A = 5 \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 15 & 5 \\ -5 & 10 \end{bmatrix}$

$7I = 7 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$

$\therefore A^2 - 5A + 7I$
 $= \begin{bmatrix} 8 & 5 \\ -5 & 3 \end{bmatrix} - \begin{bmatrix} 15 & 5 \\ -5 & 10 \end{bmatrix} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0 = \text{RHS}$

9] Find x , if $\begin{bmatrix} x & -5 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & -1 \\ 2 & 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ 4 \\ 1 \end{bmatrix} = 0$?

Sol] LHS = $\begin{bmatrix} x-2 & -10 & 2x-8 \end{bmatrix} \begin{bmatrix} x \\ 4 \\ 1 \end{bmatrix} = \begin{bmatrix} x^2-2x+(-40)+2x-8 \end{bmatrix}$

$x^2 - 2x - 40 + 2x - 8 = 0$

$x^2 - 48 = 0$

$x = \sqrt{48}$

$x = \pm 4\sqrt{3}$

11] Find matrix X so that $X \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = \begin{bmatrix} -7 & -8 & -9 \\ 2 & 4 & 6 \end{bmatrix}$

Sol] order $(X) = m \times n$

order $(Y) = 2 \times 3$

order $(Z) = 2 \times 3$

$\therefore (m \times n)(2 \times 3) = (2 \times 3)$

According to multiplication of matrix

$m = 2$ and

$n = 2$

$5 = 6s + t$

$0 = 6s$

$0 = 6t$

ACHIEVER

$$\begin{aligned}
 \therefore X &= \begin{bmatrix} a & b \\ c & d \end{bmatrix} \\
 \Rightarrow \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} &= \begin{bmatrix} -7 & -8 & -9 \\ 2 & 4 & 6 \end{bmatrix} \\
 \begin{bmatrix} a+4b & 2a+5b & 3a+6b \\ c+4d & 2c+5d & 3c+6d \end{bmatrix} &= \begin{bmatrix} -7 & -8 & -9 \\ 2 & 4 & 6 \end{bmatrix} \\
 \Rightarrow a+4b &= -7 \rightarrow (1) \\
 c+4d &= 2 \rightarrow (2) \\
 2a+5b &= -8 \rightarrow (3) \\
 2c+5d &= 4 \rightarrow (4) \\
 \text{Consider } (3) - (1) & \\
 2a+5b &= -8 \\
 a+4b &= -7 \\
 \hline
 a+b &= -1 \rightarrow (5) \\
 a &= -1-b \rightarrow (5) \\
 \text{Substitute (5) in (1)} & \\
 -1-b+4b &= -7 \\
 -1+3b &= -7 \\
 3b &= -6 \\
 b &= -2 \rightarrow (6) \\
 \text{Substitute (6) in (5)} & \\
 a &= -1-(-2) \\
 a &= 1 \\
 \text{Consider } (4) - (2) & \\
 2c+5d &= 4 \\
 c+4d &= 2 \\
 \hline
 c+d &= 2 \\
 c &= 2-d \rightarrow (7) \\
 \text{Substitute (7) in (2)} & \\
 2-d+4d &= 2 \\
 2+3d &= 2 \\
 3d &= 0 \\
 d &= 0 \rightarrow (8)
 \end{aligned}$$

Substitute ③ in ⑦

$$C = 2 - 0$$

$$\boxed{C = 2}$$

$$\therefore X = \begin{bmatrix} 1 & -2 \\ 2 & 0 \end{bmatrix}$$

Eg 25] Find P^{-1} if it exists, given $P = \begin{bmatrix} 10 & -2 \\ -5 & 1 \end{bmatrix}$

Sol] $|P| = 10 - 10 = 0$
 $\Rightarrow P$ is singular matrix
 $\therefore P^{-1}$ does not exist.

Eg 24] Obtain the inverse of $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$

Sol] $A = I A$

$$\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$R_2 \leftrightarrow R_1$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 3 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$R_3 \rightarrow R_3 - 3R_2$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & -5 & -8 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & -3 & 1 \end{bmatrix} A$$

$R_1 \rightarrow R_1 - 2R_2$

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & -5 & -8 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & -3 & 1 \end{bmatrix} A$$

Achiever

$$R_3 \rightarrow R_3 + 5R_2$$

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 \\ 1 & 0 & 0 \\ 5 & -3 & 1 \end{bmatrix} A$$

$$R_3 \rightarrow R_3 \div 2$$

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 \\ 1 & 0 & 0 \\ \frac{5}{2} & -\frac{3}{2} & \frac{1}{2} \end{bmatrix} A$$

$$R_2 \rightarrow R_2 - 2R_3$$

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 & 0 \\ -4 & 3 & -1 \\ \frac{5}{2} & -\frac{3}{2} & \frac{1}{2} \end{bmatrix} A$$

$$R_1 \rightarrow R_1 + R_3$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -4 & 3 & -1 \\ \frac{5}{2} & -\frac{3}{2} & \frac{1}{2} \end{bmatrix} A$$

$$I = BA$$

$$A^{-1} = B = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -4 & 3 & -1 \\ \frac{5}{2} & -\frac{3}{2} & \frac{1}{2} \end{bmatrix}$$

Ex 11] 2 farmers R & S cultivate 3 varieties of rice namely A, B, C. The sale of these varieties by both in month of Sept & Oct are given by the following matrices.

$$\text{Sept sales in ₹} \quad X = \begin{bmatrix} A & B & C \\ 10000 & 20000 & 30000 \\ 50000 & 30000 & 10000 \end{bmatrix} \begin{matrix} R \\ S \end{matrix} \quad \text{Oct sales in ₹} \quad Y = \begin{bmatrix} A & B & C \\ 5000 & 10000 & 6000 \\ 20000 & 10000 & 10000 \end{bmatrix} \begin{matrix} R \\ S \end{matrix}$$

(i) Find the combined sales in Sept & Oct for each farmer in each variety.

$$\text{Ans: Combined sales} = X + Y = \begin{bmatrix} A & B & C \\ 15000 & 30000 & 36000 \\ 70000 & 40000 & 20000 \end{bmatrix} \begin{matrix} R \\ S \end{matrix}$$

(ii) Find the decrease in sales from Sept to Oct.

$$\text{Ans: Decrease in sales} = X - Y = \begin{bmatrix} 5000 & 10000 & 24000 \\ 30000 & 20000 & 0 \end{bmatrix}$$

(iii) If both farmers receive 2% profit, compute the profit for each farmer, for each variety sold in Oct.

$$2\% = \frac{2}{100} = 0.02$$

$$\text{Profit obtained from Oct sales} = 0.02(Y)$$

$$= 0.02 \begin{bmatrix} 5000 & 10000 & 6000 \\ 20000 & 10000 & 10000 \end{bmatrix}$$

$$= \begin{bmatrix} 100 & 200 & 1200 \\ 400 & 200 & 200 \end{bmatrix}$$

Eg19] In a LA election, A political group hired a PRF to promote its candidates in 3 ways - telephone, house calls, letters. The cost per contact in paise is given in matrix A as $A = \begin{bmatrix} 40 \\ 100 \\ 50 \end{bmatrix}$ cost per contact
 telephone
 house call
 letter

The no. of contacts of each type made in two cities X & Y is given by $B = \begin{bmatrix} 1000 & 500 & 5000 \\ 3000 & 1000 & 10000 \end{bmatrix}$
 telephone house call letter
 X Y

Find the total amount spent by the group in 2 cities X & Y.

$$\text{Total money spent} = BA$$

$$= \begin{bmatrix} 1000 & 500 & 5000 \\ 3000 & 1000 & 10000 \end{bmatrix} \begin{bmatrix} 40 \\ 100 \\ 50 \end{bmatrix}$$

$$= \begin{bmatrix} 340000 \\ 720000 \end{bmatrix}$$

$$\text{Amount spent on X} = 340000 \text{ paise}$$

$$Y = 720000 \text{ paise.}$$

ACHIEVER

Theorem 1: For any square matrix A with real no. entries,
 $A + A'$ is a symmetric matrix and,
 $A - A'$ is a skew symmetric matrix

(a) TPT: $P = A + A'$ is symmetric $\Rightarrow P' = P$

$$\begin{aligned} P' &= (A + A')' \quad [(A + B)' = A' + B'] \\ &= A' + (A')' \quad [(A')' = A] \\ &= A' + A \end{aligned}$$

$$P' = A + A' \quad (A + B = B + A)$$

$$P' = P$$

$\therefore P$ is symmetric

i.e., $A + A'$ is symmetric.

(b) $Q = A - A'$ is skew symmetric

i.e., $Q' = -Q$

$$Q' = (A - A')'$$

$$Q' = A' - (A')'$$

$$= A' - A$$

$$= -(A - A')$$

$$Q' = -Q$$

Theorem 2: Any square matrix can be expressed as the sum of a symmetric and a skew symmetric matrix.

**

Theorem 3: (Uniqueness of Inverse): Inverse of a square matrix if it exists is unique.

Proof: Let $A = [a_{ij}]$ be a square matrix of order n

If possible let B & C be two inverses of A .

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TPT : $B = C$

Since B is inverse of A : $AB = BA = I \rightarrow (1)$

Since C is inverse of A : $AC = CA = I \rightarrow (2)$

$B = BI$

$= B(AC)$ [from (2)]

$= (BA)C$ (Associative axiom)

$B = IC$

$B = C$

$\Rightarrow A$ has unique inverse.

**** Theorem 4:** If A and B are invertible matrices of same order, then $(AB)^{-1} = B^{-1}A^{-1}$

TPT : $(AB)^{-1} = B^{-1}A^{-1}$

Proof : $(AB)(AB)^{-1} = I$

• Premultiply by A^{-1} on both sides

$A^{-1}(AB)(AB)^{-1} = A^{-1}I$

$(A^{-1}A)B(AB)^{-1} = A^{-1}$

$I B(AB)^{-1} = A^{-1}$

$B(AB)^{-1} = A^{-1}$

• Premultiply by B^{-1} on both sides

$(B^{-1}B)(AB)^{-1} = B^{-1}A^{-1}$

$I(AB)^{-1} = B^{-1}A^{-1}$

$(AB)^{-1} = B^{-1}A^{-1}$

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